

CATBALOGAN, Philippines

WMO: 985480

Lat: 11.7751N

Lon: 124.8842E

Elev: 5

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.4	23.1	20.0	14.7	24.5	20.8	15.5	25.2	7.1	26.5	4.8	27.1	0.5	40	0.319

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.8	34.2	27.3	33.7	27.2	33.2	27.2	28.4	32.5	28.1	32.3	27.7	31.8	2.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	31.1	26.9	22.5	30.8	26.6	22.2	30.5	91.6	32.8	89.8	32.4	88.7	32.2	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.1	4.2	3.5	DB	19.3	35.9	2.5	1.7	17.5	37.1	16.0	38.1	14.6	39.1	12.8	40.4	
			WB	18.7	29.9	2.1	1.0	17.2	30.7	15.9	31.2	14.7	31.8	13.2	32.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	26.7	27.0	27.8	28.8	29.4	29.0	28.7	29.1	28.8	28.4	27.8	27.3
	DBStd	1.46	1.23	1.26	1.20	1.26	1.05	1.18	1.16	1.26	1.15	1.08	1.17	1.22
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6660	517	477	552	563	602	571	581	593	563	570	534	537
	CDD18.3	3618	258	244	294	313	343	321	322	335	313	312	284	279
	CDH23.3	39889	2226	2244	3030	3666	4288	3859	3781	4099	3689	3425	2916	2667
	CDH26.7	14867	546	673	1049	1503	1909	1605	1491	1763	1457	1218	918	734
Wind	WSAvg	1.3	1.1	1.1	1.2	1.2	1.3	1.3	1.4	1.5	1.5	1.2	1.0	1.3
Precipitation	PrecAvg	2555	210	130	114	102	178	204	249	250	232	291	300	294
	PrecMax	3420	644	240	390	215	369	346	414	634	519	597	564	575
	PrecMin	1661	9	10	3	33	57	83	83	101	105	104	166	140
	PrecStd	459	177	73	112	60	102	94	93	156	127	152	117	137
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	33.2	33.9	35.0	35.0	34.6	33.9	33.9	33.7	33.7	33.4	32.7
		MCWB	26.0	26.2	26.3	26.8	27.7	27.7	27.5	27.4	27.7	27.2	27.1	26.8
	2%	DB	31.3	32.2	32.9	34.1	34.2	33.8	33.1	33.2	33.0	32.9	32.5	31.9
		MCWB	26.0	26.1	26.2	26.9	27.4	27.7	27.3	27.2	27.3	27.1	26.9	26.7
	5%	DB	30.4	31.3	32.1	33.3	33.5	33.0	32.4	32.7	32.4	32.1	31.7	31.1
		MCWB	26.0	25.9	26.2	26.9	27.4	27.5	27.2	27.1	27.2	27.0	26.8	26.6
	10%	DB	29.6	30.3	31.3	32.5	32.8	32.3	31.7	32.0	31.7	31.3	30.7	30.2
		MCWB	25.7	25.7	26.1	26.8	27.4	27.3	27.0	27.0	27.0	26.9	26.7	26.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.3	27.6	27.8	28.5	29.0	28.7	28.3	28.5	28.4	28.1	28.0	27.9
		MCDB	30.5	31.7	31.9	33.0	33.4	33.0	32.4	32.5	32.4	32.0	31.8	31.1
	2%	WB	26.7	26.9	27.2	27.8	28.2	28.1	27.8	27.9	27.8	27.7	27.5	27.3
		MCDB	29.9	30.6	31.2	32.2	32.6	32.5	31.7	31.9	31.8	31.5	31.1	30.5
	5%	WB	26.4	26.5	26.8	27.5	27.9	27.7	27.5	27.6	27.5	27.3	27.1	27.0
		MCDB	29.4	30.0	30.7	31.8	32.2	32.0	31.3	31.5	31.3	31.0	30.7	30.2
	10%	WB	26.0	26.0	26.4	27.1	27.5	27.4	27.2	27.2	27.2	27.0	26.8	26.6
		MCDB	28.8	29.4	30.1	31.4	31.7	31.5	30.9	31.0	30.8	30.5	30.1	29.5
Mean Daily Temperature Range	5% DB	MDBR	5.5	6.5	6.9	7.2	6.8	6.4	5.9	5.7	5.7	6.0	6.0	5.5
		MCDBR	7.0	8.2	8.2	8.7	7.8	7.4	7.0	6.4	6.7	7.1	7.5	7.0
	5% WB	MCWBR	3.2	3.6	3.3	3.2	3.0	2.9	2.8	2.5	2.7	2.9	3.2	3.1
		MCWBR	6.1	7.0	7.2	7.6	7.2	7.0	6.5	6.1	6.2	6.5	6.6	6.2
Clear-Sky Solar Irradiance	taub		0.458	0.436	0.432	0.425	0.424	0.423	0.426	0.448	0.439	0.447	0.440	0.457
		taud	2.298	2.364	2.390	2.408	2.434	2.454	2.444	2.365	2.389	2.366	2.386	2.308
	Ebn at Noon		846	881	889	885	871	862	862	854	870	860	857	836
		Edh at Noon	130	126	125	122	116	113	114	126	123	124	118	127
All-Sky Solar Radiation	RadAvg		3.85	4.76	5.46	6.01	5.81	5.22	4.88	5.09	4.94	4.69	4.30	3.81
	RadStd		0.54	0.60	0.61	0.41	0.45	0.30	0.38	0.30	0.44	0.33	0.39	0.35

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page